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RESEARCH ARTICLE

Heisenberg Group-Based Digital Signatures for Secure Communications

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ABSTRACT This paper proposes a new digital signature algorithm that employs the Heisenberg group over a prime field and exploits its structure to provide strong cryptographic properties. The developed scheme involves the Heisenberg group within a commutator-based authentication system contributing to key exchange while providing secure means of authentication as well as data integrity. The proposed method for digital signature generation and verification is described by a comprehensive algorithm, which proof of correctness is provided by a confirmation theorem. The security analysis establishes the strength of the scheme against different cryptanalytic attacks and proves the effectiveness of non-repudiation and message integrity in the digital environment. The efficiency of the presented security scheme is also evaluated in practice, using the finite field $\mathbb{Z}_{11}$, where it is measured against other standards such as ECDSA and RSA. While it has a bigger signature size, the scheme is still efficient when it comes to signing and verifying messages which highlights its suitability for protecting communications under post-quantum cryptography. It supports the on-going progress of cryptography, giving added security to new technologies when faced with the threat of complex attacks.

INDEX TERMS Heisenberg group, digital signature, hashing, cryptographic verification, group-based cryptography, message authentication.

I. INTRODUCTION

A digital signature functions as a method of authentication, allowing the author of a message to attach a code that serves as a signature for communication. It demonstrates the objectives of authenticity, integrity, and non-repudiation by functioning as a handwritten signature. Handwritten signatures function as a means of verifying the authenticity of paper documents. Digital signatures can also be employed to verify electronic documents [1]. The framework of digital signatures is designed to create an effective operational domain. It focusses on identifying complex challenges within this structure, formulating a one-way function associated with these intricate problems, and ultimately developing an

algorithm to address them. This method tackles complex issues that are directly linked to number theory, rather than group theory. In 1976, Diffie and Hellman presented public key cryptography (PKC) for the first time. PKC seeks to streamline symmetric-key cryptography by eliminating the need for two separate keys, thus enhancing secure user interactions. The number of keys required for symmetric-key cryptography escalates quickly as the user base exchanging sensitive information expands. In PKC, users create two distinct keys: one for ensuring confidentiality and the other for facilitating public access. In encryption, the key that is made available to the public is referred to as the “public key,” whereas the key utilized for decryption is identified as the “private key.” This method, in contrast to symmetric key cryptography, operates without the necessity of a shared key. PKC’s recent notable growth in this area. The current

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approaches mostly focus on solving discrete logarithm and prime factorization problems at the same time; nevertheless, each approach has unique drawbacks and restrictions. Addressing several intricate theoretical issues is relevant to contemporary cryptography. The basic problem's continued security determines how reliable these algorithms are. The key agreement procedure was used by Mo and Chen [2] to elucidate a secure transmission technique. To transmit a message within this framework, each participant must generate a unique private key. It is crucial that both parties verify the message's legitimacy after the algorithm is finished. A technique that makes use of two complex mathematical problems prime factorizations and discrete logarithm was presented by Vishnoi and Shrivastava [1]. Various algorithms address a particular complex problem, such as the discrete logarithm problem or prime factorization. They explained the precise restrictions and disruptions present in the previous methods. Their explanation made clear that solving both problems at the same time is necessary for this strategy to be practically addressed. The algorithm, which was created from several complex problems, is not feasible. The conjugacy problem in finitely produced free groups was solved by Kim [3]. To get around the ECDSA, several authors developed this method and put in place a backdoor mechanism. Although the quantity of elliptic curves accessible via this approach may differ, it is crucial to define the parameters for every elliptic curve [5]. Khatoon et al. [6] came up with a plan that uses NKAP-ECDH-SET to increase security. They also described the message's minimal processing cost and showed how the message could be verified. A cloud-based data storage architecture was presented by Mai and Khalil [7] with the goal of protecting user privacy regarding smart meter data. This was achieved by using a homomorphic cryptosystem to encrypt the smart meter data before it was stored in the cloud. Reddy et al. have created a three-factor authenticated key agreement mechanism that ensures privacy [8], for use in client-server environments. A strategy was developed using a specialised key agreement mechanism. The results show that the algorithm's execution time is shorter than that of other algorithms with a unique key. According to the literature, a published plan integrates the idea of the double conjugacy problem into the smart meter system. In both the algorithm's development and verification stages, the double conjugacy problem is essential. A technique that uses elliptic curves to ensure data integrity in cloud contexts was presented by Kavin and Ganapathy [9]. The elliptic curve cryptography techniques and their applicability to Bitcoin and the Internet of Things were studied by Shuai Xiao and associates. Using the password key agreement procedure in elliptic curve cryptography, Liu et al. [10] presented a password-based safe key agreement mechanism. Initialization, registration, key authentication, and key acceptance are the four separate stages that make up the algorithm's structure. An Arithmetic Technique for Non-Abelian Group

Cryptosystem was the title of an article presented by Iswariya and Rishivarman [11]. In this paper, a cryptosystem based on finite non-Abelian groups is developed. Its encryption and decryption procedures rely on non-commutative groups, demonstrating a notable speedup over other important public cryptosystems. To improve the security model of AKE and protect against lost-smart-card attacks, Xie et al. [12] suggest a unique dynamic ID-based anonymous two-factor AKE protocol. Madhusudhan and Hegde [13] and Wang and Wang [14] investigated the drawbacks of Ding's dynamic ID-based remote user authentication system. Ding's plan is more expensive to implement than alternative strategies. To address the problems, a plan was presented that successfully reduced Ding's approach's shortcomings while also lowering computational expenses. A solution to the conjugacy problem in generalized Heisenberg groups was proposed by Blaney and Nikolaev [15]. The development of group theoretic cryptology attacks using hyper heuristics was described in detail by Craven and Woodward [16]. This approach provides solutions to cryptography issues within groups, making it applicable to comparable challenges. An inventive signature system based on a non-Abelian group inside a group ring was presented by Pandey and Gupta [17]. Its security depends on the difficulties posed by factorization and discrete logarithm problems. There are several facets to the conjugacy problem in non-commutative algebraic structures. Metaheuristic optimization algorithms show recent advancements which prove effective for solving complex computational problems spanning multiple subject areas. The article by Khaleel [18] shows how metaheuristic algorithms transform optimization tasks while explaining their use in choosing cryptographic parameters for improved system performance. The Teaching-Learning-Based Optimization (TLBO) algorithm which represents a bio-inspired approach enables resource allocation optimization and computational workflow optimization as explained in Abualigah et al.'s work [19] proves especially relevant for cryptographic protocol key generation and verification phases. The article by Rezk et al. [20] demonstrates how metaheuristic algorithms tackle large-scale engineering challenges which shows their transferable features to cryptography field. The growth of cryptographic research emphasizes implementing optimization algorithms to enhance security-resistant cryptographic primitives including the signature scheme based on Heisenberg groups demonstrated in this approach while reducing processing requirements. Digital signatures built with the non-commutative Heisenberg group are better cryptographically than traditional schemes relying on group structures. Kumar et al. [21] put forward a group ring-based signature scheme based on commutative groups, although it is at risk of quantum attacks, mainly by using Shor's algorithm. In this case, since the Heisenberg group is non-commutative, it is more resistant to attacks and using the Hard Commutator Inverse Problem (HCIP) would be difficult as computers could not solve this complex problem in a quantum

setting. Yin et al. [22] go on to point out digital signature threats in quantum-security environments which show that non-commutative groups play a crucial role in safeguarding data from quantum risks. Xiong et al. [23] also looked at how ECDSA is applied in blockchain systems and stated that it is not very effective when dealing with failures and growing numbers of users. The difficult group operations in the Heisenberg group improve security which is why it works well for technologies that need more secure cryptography.

PRELIMINARY CONCEPTS

II. HASH FUNCTION

A hash function takes an input message and produces a corresponding hash code or hash value as output. When a hash function is applied to inputs of arbitrarily finite lengths, it generates an output of a constant size. Three fundamental properties render a hash function secure and desirable:

- For any hash output h , it is extremely difficult to determine an input message x that satisfies the equation $H(x) = h$. This property is referred to as *one way - ness*.
- The hash of two distinct messages must not produce identical results in digital signatures. This indicates that discovering $x \neq y$ such that $H(x)$ equals $H(y)$ is computationally impractical. This characteristic is referred to as *weak collision resistance*.
- Identifying a pair (x, y) such that $H(x) = H(y)$ presents significant computational challenges. This property is characterised by *robust collision resistance*. Hash functions serve as a crucial application within the realm of cryptography, specifically referred to as *data integrity*.

III. HEISENBERG GROUP

Let F be any field such that, we may define m -dimensional Heisenberg group as upper triangular matrices as follows:

$$H^m = \left\{ \begin{bmatrix} 1 & \alpha_1 & \alpha_2 & \dots & \alpha_m & \beta \\ 0 & 1 & 0 & \dots & 0 & \gamma_1 \\ 0 & 0 & \ddots & 0 & \vdots & \vdots \\ \vdots & \vdots & 0 & \ddots & 0 & \gamma_{m-1} \\ 0 & 0 & \dots & 0 & 1 & \gamma_m \\ 0 & 0 & \dots & 0 & 0 & 1 \end{bmatrix}_{(m+2) \times (m+2)} \mid \alpha = (\alpha_1 \dots \alpha_m), \gamma = \begin{pmatrix} \gamma_1 \\ \vdots \\ \gamma_m \end{pmatrix} \in F^m, \beta \in F \right\}$$

The pair $(H^m, \cdot)$ forms a group since multiplication and inversion of such matrices yield matrices of the same type. If $m =$

1, for example, we get $H^1 = \left\{ \begin{bmatrix} 1 & \alpha & \beta \\ 0 & 1 & \gamma \\ 0 & 0 & 1 \end{bmatrix} \mid \alpha, \beta, \gamma \in F \right\}$.

A group of all 3×3 upper triangular matrices encompass various representations within functional spaces, where the centre exerts a nontrivial action upon it. Moreover,

the real and integer numbers are part of ring-structured elements, recognised by their respective continuous and discrete Heisenberg group. The ongoing group arises from the characterisation of quantum systems in a *one - dimensional* framework. The connection with *m - dimensional* systems is broader in this context. The multiplication of two Heisenberg matrices in three dimensions is represented as follows

$$\begin{bmatrix} 1 & \alpha & \beta \\ 0 & 1 & \gamma \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & \alpha_1 & \beta_1 \\ 0 & 1 & \gamma_1 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & \alpha + \alpha_1 & \beta + \beta_1 + \alpha\gamma_1 \\ 0 & 1 & \gamma + \gamma_1 \\ 0 & 0 & 1 \end{bmatrix},$$

and the inverse of the element, that is, $\begin{bmatrix} 1 & \alpha & \beta \\ 0 & 1 & \gamma \\ 0 & 0 & 1 \end{bmatrix}^{-1} =$

$$\begin{bmatrix} 1 & -\alpha & \alpha\gamma - \beta \\ 0 & 1 & -\gamma \\ 0 & 0 & 1 \end{bmatrix}.$$

The Heisenberg neutral element of group is the identity matrix. The discrete Heisenberg group x , y and z generators are the non-Abelian group of ring elements

$$\text{on the integers } \alpha, \beta \text{ and } \gamma. \text{ Such that, } x = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

$$y = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \text{ and } z = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \text{ are the generator}$$

with the center element. A polynomial growth rate of order 3 using *Bass's theorem* is used to generate any element through

$$x = \begin{bmatrix} 1 & \alpha & \beta \\ 0 & 1 & \gamma \\ 0 & 0 & 1 \end{bmatrix} = y^\beta z^\gamma x^\alpha.$$

The behavior of the Heisenberg group to modulo odd prime p over a finite field is called extra special group of exponents p .

It's worth noting that H^1 is isomorphic to F^3 , implying that the homomorphism, $\varphi : H^1 \rightarrow M_3(F)$, defined by

$$\varphi(\beta, \alpha, \gamma) = \begin{bmatrix} 1 & \alpha & \beta \\ 0 & 1 & \gamma \\ 0 & 0 & 1 \end{bmatrix} \text{ exists and can be used as a bridge}$$

when investigating the Heisenberg group. If a group G has a polycyclic series of length n , for a sequence of subgroups, that is, $G_1 \supseteq G_2 \supseteq \dots \supseteq G_n \supseteq G_{n+1} = \{e\}$, such that: $\forall i \in [1 \dots n]$. So, the two conditions $G_{i+1} \trianglelefteq G_i$ and G_i/G_{i+1} is cyclic holds, hence it's called polycyclic group (pc - group).

IV. COMMUTATORS

Consider $g_1, \dots, g_n$ as elements belonging to a group G . The subsequent notation for commutators will be employed: $[g_1, g_2] = g_1^{-1} g_2^{-1} g_1 g_2$. In a broader context, a simple commutator of weight $n \geq 2$ is defined recursively using the rule $[g_1, \dots, g_n] = [[g_1, \dots, g_{n-1}], g_n]$, with the convention that $[g_1] = g_1$. A practical shorthand representation is $[x, {}_n g] = [x, \underbrace{g \dots g}_n]$. For the convenience of the reader, we reiterate

the following property of commutators: $[xy, z] = [x, z]^y [y, z]$ and $[x, yz] = [x, z][x, y]^z$ for every x, y, z belonging to G .

TABLE 1. Performance comparison of digital signature schemes.

Scheme	Key Size (bits)	Signature Size (bits)	Signature Time (ms)	Verification Time (ms)
ECDSA	256	512	2.62	8.62
RSA	2048	256	11.15	1.02
Proposed	768	3840	0.14	1.87

V. DIGITAL SIGNATURE SCHEME PRESENTED OVER HEISENBERG GROUP

This section outlines the steps of the proposed technique and illustrates its application to a specific set of matrices. The methodology consists of three main phases: key generation, signature production, and signature validation. Public and private keys will be generated during the key generation phase. The signature is generated during the signature creation phase and functions as a code to secure the message via a hash function. Validating the signature is essential during the signature verification phase to access the message. This document delineates the steps that define the algorithm.

A. INITIAL SETUP

Consider a Heisenberg group $H(f_p)$, where p is prime number. Let the cryptographic hash function q from the message space M to the Heisenberg group $H(f_p)$. That is,

$$q : M \rightarrow H(f_p)$$

where $q = q(M) \subseteq H(f_p)$ is the Hash message M .

B. KEY GENERATION

Assume that Alice wants to send a message M as a document to Bob, for the purpose of authentication she signed the document in the form of digital signature. Alice first chooses two random elements x_p and y_p which belong to the group $H(f_p)$ and the centralizer of both elements contains identity of $H(f_p)$ and the power of these elements. She publishes her public key $z = [x_p, y_p]$ and x_p^{-1} is private key.

VI. SIGNATURE GENERATION AND VERIFICATION PROTOCOL

A. SIGNATURE GENERATION

To generate a digital signature for message M , Alice performs the following steps:

1. Select Random Element:

Alice selects a random element $g \in H(f_p)$, where $H(f_p)$ is the Heisenberg group over the prime field f_p .

2. Compute intermediate values.

- $h = gqg^{-1}$, where q is the hash of the message M .
- $o = gx_pqx_p^{-1}g^{-1}$, where x_p is a public key element.
- $v = [g^{-1}, x_p] = gx_p^{-1}g^{-1}x_p$, the commutator of g^{-1} and x_p .
- $s = g(x_pqx_p^{-1})$, an intermediate computation for signature integrity.

- $t = q(x_p^{-1}y_pgx_p)^{-1}y_p$, where y_p is another public key element.

3. Signature formation

The computed values (h, o, v, s, t) form the digital signature for M . This signature, along with the public key, is sent to Bob for verification.

VII. SIGNATURE VERIFICATION

In Signature verification, Bob receives Alice signature (h, o, v, s, t) , and computes the following for verification.

1. Compute Verification Values

- $f = vz$, where z is the public key derived from x_p and y_p .
- $w = st$, combining the signature components s and t .

2. Verify Authenticity

Bob verifies the signature by checking the following equation

$$ohf = w. \quad (1)$$

If the equation holds the signature is accepted as valid. Otherwise, it is rejected. To demonstrate the correctness and efficiency of the developed signature scheme an example is presented based on the certain elements of Heisenberg group over the finite field Z_{11} . Referring to the proposed model and the mentioned cryptographic operations, the example demonstrates the generation and confirms the validation of the signature.

Let g, x_p and $y_p \in H(Z_{11})$, defined as $g = \begin{bmatrix} 1 & 3 & 2 \\ 0 & 1 & 5 \\ 0 & 0 & 1 \end{bmatrix} \mod 11$, $x_p = \begin{bmatrix} 1 & 7 & 2 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix} \mod 11$ and $y_p = \begin{bmatrix} 1 & 5 & 7 \\ 0 & 1 & 8 \\ 0 & 0 & 1 \end{bmatrix} \mod 11$; and consider the message $M = \begin{bmatrix} 1 & 2 & 10 \\ 0 & 1 & 6 \\ 0 & 0 & 1 \end{bmatrix} \mod 11$. Let cryptographic hash function q map the message space M to $H(f_{11})$, defined as, $q : m_{ij} \rightarrow 2^{ij} \mod 11$. For the given message M , we have $q =$

$$q(M) = \begin{bmatrix} 2^1 & 2^2 & 2^{10} \\ 0 & 2^1 & 2^6 \\ 0 & 0 & 2^1 \end{bmatrix} \mod 11, \text{ and after simplification}$$

$$q = \begin{bmatrix} 2 & 4 & 1 \\ 0 & 2 & 9 \\ 0 & 0 & 2 \end{bmatrix} \mod 11.$$

VIII. COMPUTATIONS

1. Compute h

$$h = gqg^{-1}$$

$$= \begin{bmatrix} 1 & 3 & 2 \\ 0 & 1 & 5 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 4 & 1 \\ 0 & 2 & 9 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 8 & 2 \\ 0 & 1 & 6 \\ 0 & 0 & 1 \end{bmatrix} \mod 11,$$

$$\text{implies, } \begin{bmatrix} 2 & 4 & 8 \\ 0 & 2 & 9 \\ 0 & 0 & 2 \end{bmatrix} \mod 11.$$

2. Compute o

$$o = gx_pqx_p^{-1}g^{-1} = \begin{bmatrix} 2 & 4 & 0 \\ 0 & 2 & 9 \\ 0 & 0 & 2 \end{bmatrix} \mod 11.$$

3. Compute v

$$v = [g^{-1}, x_p] = gx_p^{-1}g^{-1}x_p = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \mod 11.$$

4. Compute s

$$s = g(x_pqx_p^{-1}) = \begin{bmatrix} 2 & 10 & 2 \\ 0 & 2 & 8 \\ 0 & 0 & 2 \end{bmatrix} \mod 11.$$

5. Compute t

$$t = q(x_p^{-1}y_pgx_p)^{-1}y_p = qx_p^{-1}g^{-1}y_p^{-1}x_py_p$$

$$= \begin{bmatrix} 2 & 9 & 4 \\ 0 & 2 & 10 \\ 0 & 0 & 2 \end{bmatrix} \mod 11.$$

6. Compute w

$$w = st = \begin{bmatrix} 4 & 5 & 2 \\ 0 & 4 & 3 \\ 0 & 0 & 4 \end{bmatrix} \mod 11.$$

7. Compute z and f

$$z = [x_p, y_p] = x_p^{-1}y_p^{-1}x_py_p = \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \mod 11,$$

and

$$f = vz = \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \mod 11.$$

8. Verify ohf

$$ohf = \begin{bmatrix} 2 & 4 & 0 \\ 0 & 2 & 9 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} 2 & 4 & 8 \\ 0 & 2 & 9 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \mod 11$$

$$= \begin{bmatrix} 4 & 5 & 2 \\ 0 & 4 & 3 \\ 0 & 0 & 4 \end{bmatrix} \mod 11 = w.$$

Since $ohf = w$. This example proves the correctness of the signature scheme.

IX. SECURITY STRENGTH OF HEISENBERG GROUP BASED SIGNATURE PROTOCOL

The Heisenberg group in public key cryptography exhibits polycyclic behavior if and only if there exists a subseries $G_1 \triangleright G_2 \triangleright \dots \triangleright G_{n+1} = \{1\}$ for the group G , where $\triangleright$ signifies variants of G in cyclic form. Using binary addition and multiplication operations on matrix or sparse matrix components, each positive integer of the n^{th} elements of Heisenberg creates an endless nonabelian structure, making the Heisenberg scheme a desirable choice for effective hardware and software implementation. This gives rise to a clear normal form right after group operations, making the group a potentially useful solution for cryptographic applications.

A. NON-COMMUTATIVE CRYPTOGRAPHIC SYSTEMS

Security enhancements are enabled by non-commutative Heisenberg group cryptography that employs polycyclic Heisenberg groups because their operation sequence influences results ($AB \neq BA$) thus blocking typical commutative structure attacks. The difficulty that Shor's algorithm encounters with this system makes it suitable for future cryptography developments. The Elliptic Curve Cryptography (ECC) relies on commutative (Abelian) groups formed by elliptic curves under point addition which satisfies $A + B = B + A$ as this property enables key exchange and digital signature protocols. ECDH key exchange benefits from scalar multiplication commute properties to maintain protected key agreements while protecting private keys. Also, the ECDSA digital signature protocol achieves fast cryptographic proof validation because of commutative properties. ECC relies on the ECDLP to secure its operations, but the commutative property makes it subject to attacks from Pollard's rho algorithm and potential quantum computing vulnerability through Shor's algorithm. The small and efficient size of ECC keys and its speed capabilities make this cryptosystem appropriate for utilization in limited resource systems. The Heisenberg group-based cryptography demands high computational power which hampers its operation in environments with limited processing resources. ECC provides adequate security for TLS/SSL, but Heisenberg group-based schemes excel at government communication and secure data systems which require superior resistance to sophisticated attacks over performance efficiency.

B. HARD COMMUTATOR INVERSE PROBLEM (HCIP)

The security of the proposed scheme relies on the HCIP in the Heisenberg group, defined as follows: Given $z = [x_p, y_p]$ and x_p , where finding y_p is computationally infeasible.

PARAMETER SIZES AND KEY LENGTHS

1) PRIME FIELD SELECTION

For 128-bit security, we recommend $p \approx 256$ (aligned with NIST P-256). This ensures resistance against brute-force and Shor's algorithm (quantum attacks). The selection of $p \approx 256$ corresponds to a 128-bit security level, which is essential for ensuring the long-term viability of cryptographic systems in a post-quantum environment.

2) KEY AND SIGNATURE SIZES

- **Private Key:** x_p^{-1} (3 elements $\alpha, \beta, \gamma \in Fp$), size = $3\log 2_p \approx 768$ bits.
- **Public Key:** $z = [x_p, y_p]$, (3 elements), size = 768 bits.
- **Signature:** (h, o, v, s, t) , (5 matrices, each with 3 elements), size = $15\log 2_p \approx 3840$ bits.

X. PERFORMANCE EVALUATION AND COMPARISON

To assess the computational efficiency of the proposed Heisenberg group-based digital signature scheme, we implemented a prototype over the finite field Z11, consistent with the example provided in Section VIII. The key generation, signature production, and verification algorithms were tested on a personal computing environment with the following configuration: Intel(R) Core(TM) i5-4210M CPU @ 2.60 GHz, 8 GB DDR3 RAM, Intel(R) HD Graphics 4600, running a 64-bit Windows 10 operating system using Python. The average runtime was recorded over multiple iterations and compared against widely used schemes including ECDSA (over NIST P-256) and RSA (2048-bit key).

Table 1 summarizes the key size, signature size, and measured computation times for signature generation and verification.

The results demonstrate that the proposed scheme exhibits competitive signing and verification performance despite involving non-commutative group operations. Notably, signature generation time is lower than both ECDSA and RSA in this prototype, and verification time is comparable to that of RSA. While the signature size is significantly larger, the trade-off is justified by enhanced cryptographic hardness through the Hard Commutator Inverse Problem (HCIP) in the Heisenberg group. This experimental validation confirms the feasibility of the proposed system and supports its potential applicability in secure environments where signature compactness is not a limiting factor.

XI. CORRECTNESS THEOREM

Statement: Let $H(f_p)$ be the Heisenberg group and let (h, o, v, s, t) denote the digital signature. The verification algorithm ensures that the signature is always valid if and only if:

$ohf = w$, Where $f = vz$, and $w = st$.

Proof: We begin by calculating the value of f and w .

- Calculations of f :

$$\begin{aligned} f &= vz = (gx_p^{-1}g^{-1}x_p)(x_p^{-1}y_p^{-1}x_py_p) \\ &= gx_p^{-1}g^{-1}y_p^{-1}x_py_p \end{aligned}$$

- Calculations of w :

$$\begin{aligned} w &= st = g(x_pqx_p^{-1})q(x_p^{-1}y_pgx_p)^{-1}y_p \\ &= gx_pqx_p^{-1}qx_p^{-1}g^{-1}y_p^{-1}x_py_p \end{aligned}$$

- Verification of $ohf = w$, as follows:

$$ohf = gx_pqx_p^{-1}g^{-1}gqg^{-1}gx_p^{-1}g^{-1}y_p^{-1}x_py_p$$

Simplifying ohf :

$$ohf = gx_pqx_p^{-1}qx_p^{-1}g^{-1}y_p^{-1}x_py_p.$$

Comparing ohf with w , we observe that both expressions are identical after simplification:

$$ohf = w.$$

The equation $ohf = w$ is valid, thereby confirming the accuracy of the signature verification method. This verifies that the suggested digital signature system utilizing the Heisenberg group is both mathematically robust and functionally precise.

XII. DATA FORGING

Assume that the forger Eve substitutes the legitimate message M with a counterfeit message M_f . Consequently, when Bob calculates $q = q(M)$, he will be unable to confirm that $w = ohf$. This equation generally does not hold unless a collision occurs in the hash function for the specific selections of (M, q) and (h, o, v, s, t) , which is improbable given our assumptions on H .

XIII. CONCLUSION

This paper makes a valuable contribution to the existing literature on cryptographic methods and opens new horizons for the development and application of group theoretic cryptography. This work proposes a new digital signature algorithm based on the Heisenberg group over a prime field that can be used to provide reliable and secure cryptographic authentication. Therefore, the proposed scheme applies to the polycyclic structure of the Heisenberg group in conjunction with the commutator-based key exchange protocol that provides improved security features such as resisting common cryptographic attacks and data forgery. The correctness of the said scheme is established by way of proofs and, therefore, it is effective for security in communication over the internet. The analysis of security shows the resistance of the algorithm regarding hash collision and proves its capability to ensure data integrity and non-repudiation under reasonable cryptographic settings. Computational gains and the scalability of the proposed scheme strengthen the argument with practical examples and fits well to applications in secure communications, cloud computing, and IoT.

CONFLICTS OF INTERESTS

The author declares that she has no conflicts of interest.

DATA AVAILABILITY

All data generated or analyzed during this study are included in this published article.

REFERENCES

- [1] S. Vishnoi and V. Shrivastava, "A new digital signature algorithm based on factorization and discrete logarithm problem," *Int. J. Comput. Trends Technol.*, vol. 3, no. 4, pp. 653–657, 2012.
- [2] J. Mo and H. Chen, "A lightweight secure user authentication and key agreement protocol for wireless sensor networks," *Secur. Commun. Netw.*, vol. 2019, pp. 1–17, Dec. 2019.
- [3] S. W. Kim, "The twisted conjugacy problem for finitely generated free groups," *J. Pure Appl. Algebra*, vol. 220, no. 4, pp. 1281–1293, Apr. 2016.
- [4] S. Xiao, H. Wang, and J. Zhang, "New digital signature algorithm based on ECC and its application in Bitcoin and IoT," *Int. J. High Perform. Syst. Archit.*, vol. 10, no. 1, pp. 20–31, 2021.
- [5] G. Shankar, L. H. Ai-Farhani, P. A. C. Angelin, P. Singh, A. Alqahtani, A. Singh, G. Kaur, and I. A. Samori, "Improved multisignature scheme for authenticity of digital document in digital forensics using edward-curve digital signature algorithm," *Secur. Commun. Netw.*, vol. 2023, Apr. 2023, Art. no. 2093407.
- [6] S. Khatoon, S. M. M. Rahman, M. Alrubaian, and A. Alamri, "Privacy-preserved, provable secure, mutually authenticated key agreement protocol for healthcare in a smart city environment," *IEEE Access*, vol. 7, pp. 47962–47971, 2019.
- [7] V. Mai and I. Khalil, "Design and implementation of a secure cloud-based billing model for smart meters as an Internet of Things using homomorphic cryptography," *Future Gener. Comput. Syst.*, vol. 72, pp. 327–338, Jul. 2017.
- [8] A. G. Reddy, A. K. Das, V. Odelu, A. Ahmad, and J. S. Shin, "A privacy preserving three-factor authenticated key agreement protocol for client–server environment," *J. Ambient Intell. Humanized Comput.*, vol. 10, no. 2, pp. 661–680, Feb. 2019.
- [9] B. P. Kavin and S. Ganapathy, "A new digital signature algorithm for ensuring the data integrity in cloud using elliptic curves," *Int. Arab J. Inf. Technol.*, vol. 18, no. 2, pp. 180–190, 2021.
- [10] P. Liu, S. H. Shirazi, W. Liu, and Y. Xie, "PKAS: A secure password-based key agreement scheme for the edge cloud," *Secur. Commun. Netw.*, vol. 2021, pp. 1–10, Oct. 2021.
- [11] S. Iswariya and A. R. Rishivarma, "An arithmetic technique for non-abelian group cryptosystem," *Int. J. Comput. Appl.*, vol. 161, no. 2, pp. 32–35, Mar. 2017.
- [12] Q. Xie, D. S. Wong, G. Wang, X. Tan, K. Chen, and L. Fang, "Provably secure dynamic ID-based anonymous two-factor authenticated key exchange protocol with extended security model," *IEEE Trans. Inf. Forensics Security*, vol. 12, no. 6, pp. 1382–1392, Jun. 2017.
- [13] R. Madhusudhan and M. Hegde, "Security bound enhancement of remote user authentication using smart card," *J. Inf. Secur. Appl.*, vol. 36, pp. 59–68, Oct. 2017.
- [14] D. Wang and P. Wang, "Offline dictionary attack on password authentication schemes using smart cards," in *Information Security (Lecture Notes in Computer Science; Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*, vol. 7807. Cham, Switzerland: Springer, 2015, pp. 221–237.
- [15] K. R. Blaney and A. Nikolaev, "A PTIME solution to the restricted conjugacy problem in generalized Heisenberg groups," *Groups Complex. Cryptol.*, vol. 8, no. 1, pp. 69–74, Jan. 2016.
- [16] M. J. Craven and J. R. Woodward, "Evolution of group-theoretic cryptology attacks using hyper-heuristics," *J. Math. Cryptol.*, vol. 16, no. 1, pp. 49–63, Oct. 2021.
- [17] A. Pandey and I. Gupta, "A new undeniable signature scheme on general linear group over group ring," *J. Discrete Math. Sci. Cryptogr.*, vol. 25, no. 5, pp. 1261–1273, Jul. 2022.
- [18] I. M. Khaleel, "Metaheuristic algorithms in optimization and its application," *Math. Appl.*, vol. 13, no. 1, p. 131, 2024.
- [19] L. Abualigah, E. Abu-Dalhoum, A. M. Ikotun, R. A. Zitar, A. R. Alsoud, N. Khodadadi, A. E. Ezugwu, E. S. Hanandeh, and H. Jia, "Teaching-learning-based optimization algorithm: Analysis study and its application," in *Metaheuristic Optimization Algorithms*. San Mateo, CA, USA: Morgan Kaufmann, 2024, pp. 59–71.
- [20] H. Rezk, A. G. Olabi, T. Wilberforce, and E. T. Sayed, "Metaheuristic optimization algorithms for real-world electrical and civil engineering application: A review," *Results Eng.*, vol. 23, Sep. 2024, Art. no. 102437.
- [21] S. Kumar, G. Mittal, and S. Kumar, "Digital signature schemes based on group ring," *Social Netw. Comput. Sci.*, vol. 3, no. 5, p. 398, Jul. 2022.
- [22] H.-L. Yin, Y. Fu, C.-L. Li, C.-X. Weng, B.-H. Li, J. Gu, Y.-S. Lu, S. Huang, and Z.-B. Chen, "Experimental quantum secure network with digital signatures and encryption," *Nat. Sci. Rev.*, vol. 10, no. 4, Mar. 2023, Art. no. nwac228.
- [23] H. Xiong, C. Jin, M. Alazab, K.-H. Yeh, H. Wang, T. R. Gadekallu, W. Wang, and C. Su, "On the design of blockchain-based ECDSA with fault-tolerant batch verification protocol for blockchain-enabled IoMT," *IEEE J. Biomed. Health Informat.*, vol. 26, no. 5, pp. 1977–1986, May 2022.

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